Lecture 7

Canonical field observables

An oscillator has a position and a conjugate momentum. A scalar field has one such pair at every spatial point. We give this statement a precise meaning by integrating fields against test functions and defining a bracket on differentiable functions of the initial data.

Objectives. *We define functional derivatives on rapidly decreasing data, prove the field bracket's Jacobi identity and recover Klein–Gordon evolution from a Hamiltonian. We identify the charge generating a spatial translation.*

By the end of this lecture you will be able to:

- Compute functional derivatives with their boundary terms.
- Verify closure and Jacobi for the stated class of functionals.
- Recover the field equation and a translation from their generators.

7.1 Functions of field data

Definition 7.1 (Schwartz phase space). The real *Schwartz space* $\mathcal{S} = \mathcal{S}(\mathbb{R}^d, \mathbb{R})$ consists of smooth functions whose seminorms

$$p_{N,\alpha}(f) = \sup_{x\in\mathbb{R}^d} (1 + |x|)^N |\partial^\alpha f(x)|$$

are finite for every integer $N \geq 0$ and every derivative multi-index α. The *Schwartz phase space* is $\mathcal{Z} = \mathcal{S} \times \mathcal{S}$, with elements $z = (\phi, \pi)$. Convergence means convergence in every seminorm of both components.

The field ϕ and its momentum π are data on one fixed time slice. For $h = (h_\phi, h_\pi) \in \mathcal{Z}$, use the real L^2 pairing

$$(z, h)_0 = \int_{\mathbb{R}^d} (\phi h_\phi + \pi h_\pi)\, \mathrm{d}^d x.$$

It is continuous in the Schwartz topology by Cauchy–Schwarz and the decay seminorms. This topology also makes differentiation and multiplication continuous, by the product rule for derivatives.

A *functional* is a map from $\mathcal{Z}$ to $\mathbb{R}$. Here a map is *smooth* if its iterated directional derivatives of all orders exist, are multilinear in the directions, and are jointly continuous in the base point and directions, using the stated topologies. For example, its first derivative is

$$DF(z)h = \frac{\mathrm{d}}{\mathrm{d}\varepsilon} F(z + \varepsilon h)\Big|_{\varepsilon=0}.$$

The product rule follows by restricting to affine families of variations. For the chain rule, a scalar or Schwartz-valued smooth map $\mathcal{T}$ satisfies the straight-segment fundamental theorem

$$\mathcal{T}(z+h) - \mathcal{T}(z) = \int_0^1 D\mathcal{T}(z+sh)h \, \mathrm{d}s.$$

For a Schwartz-valued integrand, the integral is the limit of Riemann sums in every seminorm. Continuity on the compact parameter interval makes these sums Cauchy in each seminorm. Schwartz space is complete: the uniform limits of all weighted derivatives of a Cauchy sequence remain compatible derivatives, by the ordinary fundamental theorem of calculus. The limit therefore lies in Schwartz space. Evaluating at each spatial point proves the displayed identity from the scalar fundamental theorem.

For a smooth curve $z(t)$, take $\Delta t \in \mathbb{R} \setminus \{0\}$ and write $v_{\Delta t} = (z(t+\Delta t) - z(t))/\Delta t$. As $\Delta t \to 0$, this quotient converges to $\dot{z}(t)$ in every seminorm. The segment identity gives

$$\frac{\mathcal{T}(z(t+\Delta t)) - \mathcal{T}(z(t))}{\Delta t} = \int_0^1 D\mathcal{T}(z(t) + s\Delta t\, v_{\Delta t}) v_{\Delta t} \, \mathrm{d}s \longrightarrow D\mathcal{T}(z(t))\dot{z}(t).$$

Joint continuity makes this convergence uniform in $s \in [0,1]$, in every seminorm for a Schwartz-valued map. This proves the chain rule for the curves used below. Applying it to a map along each straight line gives the composition rule. Repeated differentiation proves the corresponding rules for the higher derivatives in the definition of smoothness.

Definition 7.2 (Admissible functional and functional derivative)**.** A smooth functional $F : \mathcal{Z} \to \mathbb{R}$ is *admissible* if there is a smooth map $\operatorname{grad} F : \mathcal{Z} \to \mathcal{Z}$ such that

$$DF(z)h = (\operatorname{grad} F(z), h)_0 = \int_{\mathbb{R}^d} \big(F_\phi[z]h_\phi + F_\pi[z]h_\pi\big) \, \mathrm{d}^d x \quad (h \in \mathcal{Z}). \tag{7.1}$$

Its components $F_\phi[z]$ and $F_\pi[z]$ are the *functional derivatives*. Write $\mathcal{A}$ for the class of admissible functionals.

These derivatives are unique. The difference of two candidate gradients can itself be chosen as h in (7.1), giving zero L^2 norm. Smoothness of F alone does not assert admissibility.

For a real test function $f \in \mathcal{S}$, define the *smeared fields* $\phi(f) : \mathcal{Z} \to \mathbb{R}$ and $\pi(f) : \mathcal{Z} \to \mathbb{R}$ by

$$\phi(f)[z] = \int f\phi \, \mathrm{d}^d x, \qquad \pi(f)[z] = \int f\pi \, \mathrm{d}^d x. \tag{7.2}$$

Their gradients are $(f, 0)$ and $(0, f)$, so both belong to $\mathcal{A}$. Polynomials in finitely many such smeared fields are also admissible, by the product rule. Local integrals can be admissible too. We will verify this directly for the Klein–Gordon Hamiltonian.

7.2 The field bracket and Jacobi

In mechanics, the canonical bracket pairs a position derivative with a momentum derivative. Define the fixed continuous linear map $\mathbb{J} : \mathcal{Z} \to \mathcal{Z}$ by $\mathbb{J}(a,b) = (b,-a)$. It is skew with respect to the pairing: $(a, \mathbb{J}b)_0 = -(\mathbb{J}a, b)_0$.

Definition 7.3 (Field bracket and Hamiltonian vector field)**.** For $F, G \in \mathcal{A}$, the *field bracket* is the functional

$$\{F, G\}[z] = (\operatorname{grad} F, \mathbb{J} \operatorname{grad} G)_0 = \int_{\mathbb{R}^d} (F_\phi G_\pi - F_\pi G_\phi)\, \mathrm{d}^d x. \tag{7.3}$$

The *Hamiltonian vector field* of F is the map $X_F : \mathcal{Z} \to \mathcal{Z}$ given by $X_F(z) = \mathbb{J} \operatorname{grad} F(z) = (F_\pi, -F_\phi)$.

All derivatives in (7.3) are evaluated at z. The integral is finite because its factors are Schwartz functions. The order of the bracket matters: the chain rule gives

$$X_F G := DG(z) X_F(z) = \{G, F\}[z]. \tag{7.4}$$

Proposition 7.4 (Canonical field bracket)**.** *The class $\mathcal{A}$ is closed under products and the field bracket. The bracket is bilinear and antisymmetric, obeys the product rule, and satisfies the Jacobi identity*

$$\{\{F, G\}, K\} + \{\{G, K\}, F\} + \{\{K, F\}, G\} = 0 \qquad (F, G, K \in \mathcal{A}). \tag{7.5}$$

For a smooth path with $\dot{z} = X_H(z)$, every $F \in \mathcal{A}$ obeys $\frac{\mathrm{d}}{\mathrm{d}t} F[z(t)] = \{F, H\}[z(t)]$.

Proof. We first identify the derivative needed to prove both closure and Jacobi. For $F \in \mathcal{A}$, let $H_F(z) = D(\operatorname{grad} F)(z)$. It is a continuous linear map from $\mathcal{Z}$ to $\mathcal{Z}$. Mixed ordinary derivatives of $F(z + sh + tk)$ commute. Differentiating (7.1) therefore gives the symmetry

$$(H_F h, k)_0 = D^2 F(z)[h, k] = (h, H_F k)_0 \qquad (h, k \in \mathcal{Z}). \tag{7.6}$$

This is symmetry for the real pairing; no bounded operator on all of L^2 is being asserted.

Write $a = \operatorname{grad} F$, $b = \operatorname{grad} G$ and $a_K = \operatorname{grad} K$ at the same point z. Differentiating the bracket, using (7.6) and skewness of $\mathbb{J}$, gives

$$D\{F, G\}(z)h = (H_F h, \mathbb{J} b)_0 + (a, \mathbb{J} H_G h)_0 = (h, H_F \mathbb{J} b - H_G \mathbb{J} a)_0.$$

Thus its gradient is

$$\operatorname{grad}\{F, G\} = H_F \mathbb{J} b - H_G \mathbb{J} a. \tag{7.7}$$

Both terms are Schwartz-valued and smooth. Indeed, derivatives of the smooth gradient maps are jointly smooth in their base point and direction, and their evaluation at the smooth directions $\mathbb{J} b$ and $\mathbb{J} a$ uses the chain rule. Repeated differentiation gives finite sums of higher derivatives of these gradient maps, evaluated on smooth directions. Every such term is jointly continuous by the definition of smoothness. This proves bracket closure. The gradient of a product is $F \operatorname{grad} G + G \operatorname{grad} F$, which proves product closure and the bracket product rule. Bilinearity and antisymmetry follow from the defining pairing.

For Jacobi, substitute (7.7) into the cyclic sum and collect terms having the same Hessian. The sum is

$$\begin{aligned} &(H_F \mathbb{J} b, \mathbb{J} a_K)_0 - (H_F \mathbb{J} a_K, \mathbb{J} b)_0 \\ &+ (H_G \mathbb{J} a_K, \mathbb{J} a)_0 - (H_G \mathbb{J} a, \mathbb{J} a_K)_0 \\ &+ (H_K \mathbb{J} a, \mathbb{J} b)_0 - (H_K \mathbb{J} b, \mathbb{J} a)_0 = 0. \end{aligned}$$

Each line vanishes by Hessian symmetry. Finally, differentiation along the path and (7.4) give the stated evolution of F. ∎

For the smeared fields, substitution of their constant gradients yields

$$\{\phi(f),\pi(g)\}=\int_{\mathbb{R}^d} fg\,\mathrm{d}^d x,\qquad \{\phi(f),\phi(g)\}=\{\pi(f),\pi(g)\}=0. \tag{7.8}$$

For example, in one dimension take $f(x)=e^{-x^2}$ and $g(x)=(1+x)e^{-2x^2}$. The odd term integrates to zero and the Gaussian integral gives $\{\phi(f),\pi(g)\}=\sqrt{\pi/3}$. This is the precise smeared meaning of canonical field pairs.

7.3 The Klein–Gordon Hamiltonian

Write v for the time derivative and $w\in\mathbb{R}^d$ for the spatial gradient. The Klein–Gordon density is

$$L(\phi,v,w)=\frac{1}{2}v^2-\frac{1}{2}|w|^2-\frac{1}{2}m^2\phi^2,\qquad m\geq 0.$$

Definition 7.5 (Legendre map and field Hamiltonian)**.** The *Legendre map* of this density is

$$\mathbb{R}\times\mathbb{R}\times\mathbb{R}^d\longrightarrow\mathbb{R}\times\mathbb{R}\times\mathbb{R}^d,$$
$$(\phi,v,w)\longmapsto(\phi,\pi,w),\qquad \pi=\frac{\partial L}{\partial v}.$$

Using its inverse to express v in terms of π, the *Hamiltonian density* is $\mathcal{H}=\pi v-L$. The *field Hamiltonian* is the functional $H:\mathcal{Z}\to\mathbb{R}$ given by

$$H[\phi,\pi]=\int_{\mathbb{R}^d}\mathcal{H}(\phi,\pi,\nabla\phi)\,\mathrm{d}^d x.$$

For this density the momentum is $\pi=v$, so the Legendre map is a bijection. The subtraction $\pi v-L$ gives

$$\mathcal{H}=\frac{1}{2}\bigl(\pi^2+|w|^2+m^2\phi^2\bigr),\qquad H[\phi,\pi]=\frac{1}{2}\int_{\mathbb{R}^d}\bigl(\pi^2+|\nabla\phi|^2+m^2\phi^2\bigr)\,\mathrm{d}^d x. \tag{7.9}$$

We compute its derivative, keeping the boundary term first. On the cube $Q_R=[-R,R]^d$, integration by parts gives

$$\int_{Q_R}\nabla\phi\cdot\nabla h_\phi\,\mathrm{d}^d x=\int_{\partial Q_R}h_\phi\,\partial_n\phi\,\mathrm{d}S-\int_{Q_R}(\Delta\phi)h_\phi\,\mathrm{d}^d x.$$

Schwartz decay makes the face integrals tend to zero as $R\to\infty$. For example, choose a decay exponent larger than d and bound the integrand on each face by that decay bound times its area. The derivative and its gradient are therefore

$$DH(z)h=\int\bigl[\pi h_\pi+(-\Delta\phi+m^2\phi)h_\phi\bigr]\,\mathrm{d}^d x,\qquad \operatorname{grad}H=(-\Delta\phi+m^2\phi,\pi). \tag{7.10}$$

The gradient is a continuous linear map on $\mathcal{Z}$, so $H\in\mathcal{A}$.

The *Hamilton equations* are $\dot z=X_H(z)$, namely

$$\phi_t=\pi,\qquad \pi_t=\Delta\phi-m^2\phi. \tag{7.11}$$

They are equivalent to $(\Box+m^2)\phi=0$ with $\pi=\phi_t$. Existence is supplied by the actual Fourier solution of Lecture 4. For Schwartz data it gives a smooth path in $\mathcal{Z}$: the multipliers $\cos(t\nu)$

and $\sin(t\nu)/\nu$ are smooth in k with polynomially bounded k-derivatives on each bounded time interval, as are their time derivatives. At $m = 0$ and $k = 0$, their power series in $|k|^2$ remove the apparent singularity. Away from that point the bounds follow by differentiation. The product rule and the Schwartz Fourier automorphism from *Analysis* I, Lecture 7, then give the assertion in every seminorm. Thus Hamilton evolution agrees with the unique energy solution of Lecture 4. Antisymmetry also gives $\dot H = \{H, H\} = 0$ along this path.

7.4 A charge generating translations

Lecture 3 associates a translation in direction i with the characteristic $Q^1 = \partial_i\phi$. The superscript 1 labels the single field component. Its charge density is the mixed stress component $T^0{}_i = \pi\partial_i\phi$. Its spatial translation charge $J_i : \mathcal{Z} \to \mathbb{R}$ is

$$J_i[\phi, \pi] = \int_{\mathbb{R}^d} \pi\partial_i\phi \, \mathrm{d}^d x, \qquad i = 1, \dots, d. \tag{7.12}$$

This is the charge with a lower spatial index. The physical momentum $\int T^{0i}$ has the opposite sign in our metric convention. Integration by parts retains $\int_{\partial Q_R} \pi h_\phi n_i \, \mathrm{d}S$ before Schwartz decay removes it. Hence

$$\operatorname{grad} J_i = (-\partial_i\pi, \partial_i\phi), \qquad X_{J_i} = (\partial_i\phi, \partial_i\pi).$$

The gradient is continuous linear, so this charge is admissible. Its Hamiltonian vector field is exactly the derivative at zero of $(\phi(x + \varepsilon e_i), \pi(x + \varepsilon e_i))$. In terms of admissible observables, this assertion reads

$$\{\phi(f), J_i\} = \int f\partial_i\phi \, \mathrm{d}^d x, \qquad \{\pi(g), J_i\} = \int g\partial_i\pi \, \mathrm{d}^d x. \tag{7.13}$$

The Noether charge therefore generates its characteristic, in the stated bracket order. It is conserved under Klein–Gordon evolution. Indeed, integration by parts and decay give

$$\{H, J_i\} = \int \left[(-\Delta\phi + m^2\phi)\partial_i\phi + \pi\partial_i\pi\right] \mathrm{d}^d x = \frac{1}{2}\int \partial_i\left(|\nabla\phi|^2 + m^2\phi^2 + \pi^2\right) \mathrm{d}^d x = 0.$$

Consequently $\dot J_i = \{J_i, H\} = 0$.

Corollary 7.6 (Klein–Gordon and translation generators). *For the Klein–Gordon density, the Legendre map is bijective. The functionals H and J_i belong to $\mathcal{A}$. The equation $\dot z = X_H(z)$ is the Klein–Gordon system (7.11). The vector field X_{J_i} is the infinitesimal generator of the spatial translation $(\phi, \pi) \mapsto (\phi(\,\cdot + \varepsilon e_i), \pi(\,\cdot + \varepsilon e_i))$. Both charges are conserved under Klein–Gordon evolution.*

Proof. The momentum computation gives $\pi = \nu$. The gradients in (7.10) and the translation computation above give the two vector fields. Along the smooth Klein–Gordon path constructed above, Proposition 7.4, antisymmetry and $\{H, J_i\} = 0$ prove conservation. ∎

7.5 Point values fall outside this class

Fix $x_0 \in \mathbb{R}^d$. The functional $F[z] = \phi(x_0)$ is smooth on $\mathcal{Z}$, but it is not admissible. Its derivative is $DF(z)h = h_\phi(x_0)$. If a Schwartz function a represented this derivative, it would satisfy

$$\int a(x)h(x) \, \mathrm{d}^d x = h(x_0) \qquad (h \in \mathcal{S}).$$

Tests supported away from x_0 force $a = 0$ there by the fundamental lemma. Continuity forces $a(x_0) = 0$ too. A test with $h(x_0) = 1$ gives a contradiction. The formal derivative is instead the delta distribution, defined by $\langle \delta_{x_0}, h \rangle = h(x_0)$. It is not a Schwartz function. Thus a formal pointwise bracket kernel does not enlarge the class in Proposition 7.4. Sharp spatial boundaries can also produce distributions when a functional is varied. Lectures 11 and 12 must specify the functionals and their boundary conditions before using their brackets.

Summary. On real Schwartz pairs, admissible functionals have smooth Schwartz-valued functional derivatives. Their canonical bracket is closed and satisfies Jacobi because the Hessians are symmetric and $\mathbb{J}$ is constant. The Klein–Gordon Hamiltonian gives $\phi_t = \pi$ and $\pi_t = \Delta\phi - m^2\phi$. The lower-index translation charge generates $\partial_i\phi$ and $\partial_i\pi$. Point evaluation is smooth but is excluded from this bracket class. Lecture 8 describes the corresponding pairing directly on solutions.

Self-check. Compute the functional derivatives of $F[\phi, \pi] = \int \phi^3 \, \mathrm{d}^d x$. Does F belong to the stated class?

Answer. The gradient is $(3\phi^2, 0)$. Multiplication is continuous on Schwartz space, so this gradient is a smooth Schwartz-valued polynomial map. Thus $F \in \mathcal{A}$.

Self-check. Compute $\{\pi(g), \phi(f)\}$ and $\{\phi(f), H\}$.

Answer. The values are $-\int fg \, \mathrm{d}^d x$ and $\int f\pi \, \mathrm{d}^d x$, respectively. The second is the time derivative of the smeared displacement.

Self-check. Which hypothesis makes the two terms in each line of the Jacobi computation cancel? Would skewness of a field-dependent replacement for $\mathbb{J}$ suffice?

Answer. Hessian symmetry cancels each pair. Constancy of $\mathbb{J}$ ensures that its derivative contributes no extra terms. Skewness alone does not remove such terms for a field-dependent replacement.

Self-check. Use the physical momentum $P^i = \int T^{0i} = -J_i$ as generator. Which spatial translation does it produce?

Answer. Its vector field is $-(\partial_i\phi, \partial_i\pi)$, so it generates $(\phi(x - \varepsilon e_i), \pi(x - \varepsilon e_i))$. This sign follows from raising the spatial stress index.